Assignment 2: Activity Classification

Overview

You will build an activity classifier for **seven activities** — walking, walking upstairs, walking downstairs, jogging, standing, sitting, and lying from the accelerometer data the class collected in the Activities Study. This is your introduction to the machine learning workflow: windowing, feature extraction, model training, evaluation.

Done in **groups of three**, using the class corpus from the Activities Study.

Use of AI coding assistants (Claude Code, Cursor, Copilot, ChatGPT) is **permitted** on the code and the exploration report. The assignment quiz is closed-resource and graded on whether you understood what you were doing.

What's in this repo

```
├── RUBRIC.md                       # assignment instructions and grading
├── A2_activity_recognition.ipynb   # student notebook with TODOs for features, models, aggregation, and failure analysis
├── util.py                         # magnitude, filtering, feature, and decision-tree helpers
└── data/
    ├── Class_Data/                 # class corpus as of Oct 5; 615 accelerometer CSVs
    │   ├── downstairs/             # 88 recordings
    │   ├── jogging/                # 39 recordings
    │   ├── lying/                  # 82 recordings
    │   ├── sitting/                # 118 recordings
    │   ├── standing/               # 92 recordings
    │   ├── upstairs/               # 84 recordings
    │   └── walk/                   # 112 recordings
    │                               # filenames: accelerometer_<zipfile_name>.csv
    └── Sample_Data/                # 35 sample CSVs (wrist-worn device, converted to Sensor Logger units)
        ├── downstairs/            # 5 recordings
        ├── jogging/               # 5 recordings
        ├── lying/                 # 5 recordings
        ├── sitting/               # 5 recordings
        ├── standing/              # 5 recordings
        ├── upstairs/              # 5 recordings
        └── walk/                  # 5 recordings
```

Data

A2 uses recordings from the **CS328 Activities Study** you joined in A0: 30-60 s of a single activity at a natural pace, no counting. Same sensors and CSV format as A1, so your A1 loading and filtering code could carry over. Labels come from the Activity dropdown:

- **walking** → recordings labeled "walk"
- **walking downstairs** → recordings labeled "downstairs"
- **walking upstairs** → recordings labeled "upstairs"
- **standing** → recordings labeled "standing"
- **sitting** → recordings labeled "sitting"
- **lying** → recordings labeled "lying"
- **jogging/running** → recordings labeled "jogging"

Each is in a folder in the ./data/Class_Data and ./data/Sample_Data folders.

Analysis

Your job is to build a **classification pipeline** that takes a recording, breaks it into windows, computes features, and assigns one of seven labels.

Two design axes

A2 has **two independent design choices**. Treat them separately in the report.

Axis	What you choose	Example knobs
Feature engineering	Which features per window + window length	Time-domain stats (mean, std, RMS, range, percentiles), frequency-domain (dominant freq, spectral energy, FFT-peak ratio), window length (1s/3s/5s/10s), window overlap
Model choice	Which classifier + hyperparameters	Decision Tree, Random Forest, kNN, Logistic Regression, SVM, Gradient Boosting — anything sklearn classical. (Skip deep learning at A2 — classical models on window features are the right tool here.)

Better features can let a simpler model win; a more powerful model can compensate for thin features. The report will ask you to think about both.

Required: compare at least 2 models

You must explore **at least two different classifier types** (e.g., Decision Tree vs Random Forest, kNN, Logistic Regression, etc.) and compare them on the same feature set. This makes "model choice" an explicit decision you document, not a hidden one.

Deliverables

Submit a zip file on Canvas by Oct 18, 11:59 PM, one submission per group:

1. completed `A2_activity_recognition.ipynb` notebook and any potential .py files generated by you.

2. `exploration_report.pdf` — your systematic exploration (the depth artifact)

Use the templates below. Same 5-section structure as A1 — feature exploration + model exploration replace A1's filter + peak detection.

§1: Feature exploration

Which features did you try? Which discriminated which classes? Aim for ~5-8 features in your final pipeline; the table can show more that you tried and dropped.

Feature	Class it helps discriminate	Effect on accuracy (+/− / no change)	Rationale
(e.g., dominant frequency 1-5 Hz)	walking vs jogging	+6%	Jogging cadence ~3 Hz, walking ~2 Hz; clear frequency separation
(e.g., RMS magnitude)	walking vs stairs	+3%	Stairs have higher impulse → larger RMS
…	…	…	…

§2: Model exploration

For each model you tried:

Model	Key hyperparameter(s)	Test accuracy	Notes
(e.g., Decision Tree)	max_depth=3, entropy	76%	Baseline, interpretable
(e.g., Random Forest)	n_estimators=100, max_depth=10	84%	Better but black-box; gains came from stairs class
...	...	...	...

One paragraph: which won and why? Did the better model help all classes equally, or did the gain concentrate on specific classes?

§3: Your universal pipeline

- The exact feature set + classifier + hyperparameters.
- Window length and aggregation strategy (majority vote? duration-weighted? confidence-weighted?).
- One paragraph: how you decided what to lock in.

§4: Confusion matrix + per-class analysis

Predicted ↓ / Actual →	walking	upstairs	downstairs	jogging	standing	sitting	lying
walking	_	_	_	_	_	_	_
upstairs	_	_	_	_	_	_	_
downstairs	_	_	_	_	_	_	_
jogging	_	_	_	_	_	_	_
standing	_	_	_	_	_	_	_
sitting	_	_	_	_	_	_	_
lying	_	_	_	_	_	_	_

For each off-diagonal cell that is meaningfully nonzero:

- One sentence: which feature(s) or model behavior caused this confusion?
- Attribute the confusion to one of the two design axes: **feature side** (feature set doesn't separate these classes) or **model side** (features separate them, but the model failed to use them).

§5: Top failure modes + edge cases

For each of your **top 2 failure modes**:

- The class pair (e.g., stairs-up classified as walking).
- A specific recording or window where the failure is clear. Embed a figure showing the trace + the feature values for that window vs. typical values for the correct class.
- 1-2 sentences: the physical mechanism causing the failure.
- 1 sentence: is this primarily a **feature** problem or a **model** problem?
- 1 sentence: one specific change that would help.

3. `writeup.pdf` — half-page reflection

Cover, briefly:

- What was the single most important decision your group made (feature or model side)? (~2 sentences)
- What surprised you most when you looked at the confusion matrix? (~2 sentences)
- What would you try next, if you had more time? (~1 sentence)

4. Contribution statement

One line: who did what.

Assessment

Total: 48 points. Mix of group-level (artifact) and individual (assignment quiz and data collection). The general concept quiz on activity recognition / classification is graded separately as part of the class quiz series.

Individual-level (24 pts)

- **Assignment Quiz (12 pts)**
- **Data Collection (12 pts)**

Group-level (24 pts)

- **Code grading (12 pts)**
- **Exploration report (10 pts)**
- **Writeup + contribution statement (2 pts)**

We use broad buckets, not percentile ranking, as Mobile health classification metrics are noisy. Depth of understanding is graded at the assignment quiz.

On using AI

You may use AI coding assistants on the notebook, the exploration report, and the writeup. The assignment quiz tests whether you, specifically, engaged with the work. Questions on the quiz directly reference specific sections of your group's submitted exploration report — vague answers that don't match the report will be flagged.

Late Policy

- Each group has **5 late days**, usable across all assignments for final code submissions.
- Late submission penalty: −10% per day after the 5 free late days.
- **Assignment quiz is individual; no late days apply** except for documented emergencies.
- Significant contribution imbalance (as reported) → individual grading may be triggered.

FAQ

Q: Can I use ChatGPT/Claude to write the notebook and the exploration report? Yes. You're being assessed on understanding, not typing. The quiz catches faking.

Q: Can I use deep learning (CNN, LSTM, etc.)? Skip it for A2. Hand-crafted window features + classical ML is the right tool for this class set; deep learning is overkill and the GPU/setup overhead isn't worth it. A3 will allow more powerful models if you want.

Q: How do I aggregate window predictions to a recording-level label? Majority vote is fine. You can also try duration-weighting (longer-running classes get higher weight) or confidence-weighting. Document your choice in §3.

Q: What window length should I use? Try several. A few seconds typically works for gait-based activities, but longer windows can capture rest patterns. Report what you tried in §1.

Q: Why does the baseline confuse stairs and walking? Stairs-up has a vertical impulse pattern similar to walking but with different amplitude and timing. Without features that capture vertical-axis specifics or step periodicity changes, the classifier can't tell them apart. Your job is to fix this.

Q: Why are lying, sitting, and standing so hard to separate? Sensor Logger removes gravity, so the phone's orientation (flat vs upright) is not in the data. All three look like "almost no motion" on the acceleration magnitude. Think about what is still different (e.g., body sway while standing, small movements while sitting), and expect some confusion between these classes. Explain what you see in §4.